Lab#5: Logical, Bit Manipulation and String instructions

Assignment #1: Write an assembly code to find number of ones in each number in the array X and store the result in the array Y

X DB 12H, 0F5H, 0A7H
Y DB 3 DUP(0)

Solution:

```
01 ORG 100H
02
03
04          MOV CL,03
05          MOV DL,0
06          MOV SI,-1
07 L:       INC SI
08          CLC
09 L1:      SHR X[SI],1
10          JC ONES
11          CMP X[SI],0
12          JNE L1
13          LOOP L
14 ONES:    INC Y[SI]
15          CMP X[SI],0
16          JNE L1
17          LOOP L
18
19 HLT
20    X DB 12H,0F5H,0A7H
21    Y DB 3 DUP(0)
```

Assignment #2: Write an assembly code to replace the even numbers in the array X with zero (Note: without using DIV or IDIV instructions):

X DB 10, 3, 1, 2

Solution:

```
01 ORG 100H
02
03          MOV SI,0
04 L1:      MOV AL,0
05          CMP SI,4
06          JE EXIT
07          MOV AL,X[SI]
08          SHR AL,1
09          MOV BL,2
10          MUL BL
11          CMP AL,X[SI]
12          JE ZERO
13          INC SI
14          JMP L1
15 ZERO:    MOV X[SI],0
16          INC SI
17          JMP L1
18 EXIT:
19
20 HLT
21      X DB 10,3,1,2
```

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Assignment #3: Write an assembly code to count number of a's in the following string (Note: using SCASB instruction and REPNE prefix):

STR DB 'aabcadxyaaAa'

Solution:

```
ORG 100H

        MOV CX,12
        MOV AL,'a'
        MOV DI,offset STR
        MOV DL,0
L1:     REPNE SCASB
        JNE END
        INC DL
        JCXZ END
        JMP L1
END:

HLT
    STR DB 'aabcadxyaaAa'
```

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Assignment #4: Write an assembly code to reverse the string STR1 in STR2

STR1 DB 'Assembly'
LENGTH EQU ($-STR1)
STR2 DB LENGTH DUP(0)

Solution:

```
ORG 100H


    MOV CX,8
    MOV SI,7
    MOV DI,0

L1: MOV AL,STR1[SI]
    MOV STR2[DI],AL
    INC DI
    DEC SI
    LOOP L1


HLT
    STR1 DB 'Assembly'
    LENGTH EQU ($-STR1)
    STR2 DB LENGTH DUP(0)
```

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